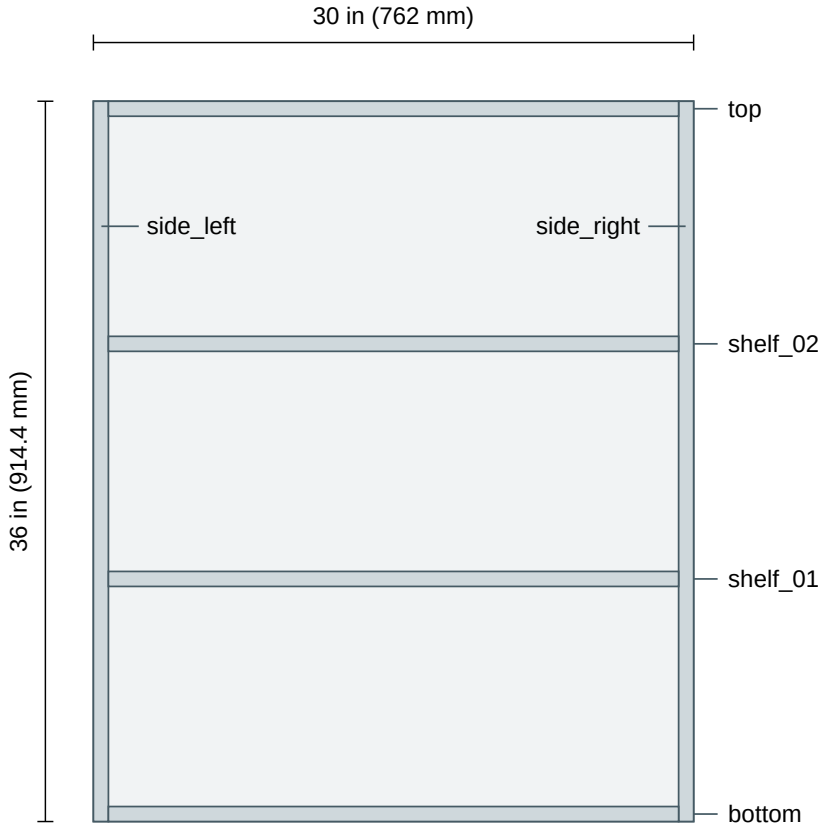
Design overview

Reference plywood bookcase

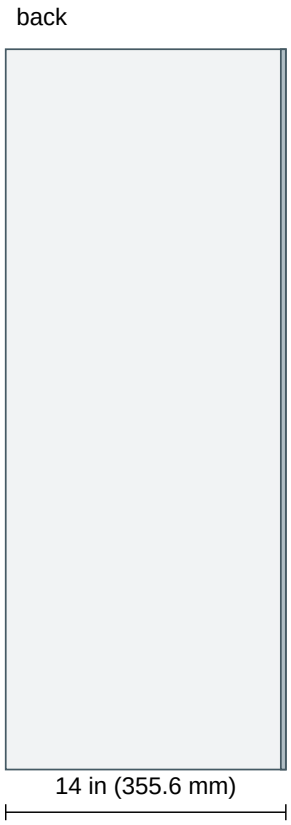
TECHNICAL FABRICATION METHOD COMPLETE - PHYSICAL REVIEW PENDING

Saved CAD, cut dimensions, drawings and this packet describe the same immutable revision. Outside depth includes the applied back. Dimensions are nominal; no fabrication fit tolerance or shelf load rating is supplied.

Verified geometry	7 physical panels; 3 semantic definitions; 3 manufacturing cut groups
Unit reference	1 in (25.4 mm); approximate fractions are explicitly labeled.



FRONT - full-height sides and between-side panels



RIGHT - back included in outside depth

Instances and clear openings

Board identity and installed centers (millimeters)

side_left	panel_side | center X 9.525, Y 174.625, Z 457.2 mm
side_right	panel_side | center X 752.475, Y 174.625, Z 457.2 mm
bottom	panel_horizontal | center X 381, Y 174.625, Z 9.525 mm
shelf_01	panel_horizontal | center X 381, Y 174.625, Z 307.975 mm
shelf_02	panel_horizontal | center X 381, Y 174.625, Z 606.425 mm
top	panel_horizontal | center X 381, Y 174.625, Z 904.875 mm
back	panel_back | center X 381, Y 352.425, Z 457.2 mm

Clear openings - measured between saved opposing faces

Opening 1	Width 28 1/2 in (723.9 mm); depth 13 3/4 in (349.25 mm); clear height 11 in (279.4 mm). Z faces 19.05 to 298.45 mm.
Opening 2	Width 28 1/2 in (723.9 mm); depth 13 3/4 in (349.25 mm); clear height 11 in (279.4 mm). Z faces 317.5 to 596.9 mm.
Opening 3	Width 28 1/2 in (723.9 mm); depth 13 3/4 in (349.25 mm); clear height 11 in (279.4 mm). Z faces 615.95 to 895.35 mm.

Spec SHA-256 e44d9f09f593efb98fe8cc4da6505196e45b01f9e4012e8ba5e391ae3b35d409

Grouped cuts and materials

Finished sizes, rough blanks and kerf remain separate. Grain is local L where declared. Groups combine panels only when their complete manufacturing fingerprint matches.

Quantity 2 - panel_side

Members: side_left, side_right

Finished L / W / T	36 in (914.4 mm) | 13 3/4 in (349.25 mm) | 3/4 in (19.05 mm)
Rough L / W / T	36 in (914.4 mm) | 13 3/4 in (349.25 mm) | 3/4 in (19.05 mm)

Quantity 4 - panel_horizontal

Members: bottom, shelf_01, shelf_02, top

Finished L / W / T	28 1/2 in (723.9 mm) | 13 3/4 in (349.25 mm) | 3/4 in (19.05 mm)
Rough L / W / T	28 1/2 in (723.9 mm) | 13 3/4 in (349.25 mm) | 3/4 in (19.05 mm)

Quantity 1 - panel_back

Members: back

Finished L / W / T	36 in (914.4 mm) | 30 in (762 mm) | 1/4 in (6.35 mm)
Rough L / W / T	36 in (914.4 mm) | 30 in (762 mm) | 1/4 in (6.35 mm)

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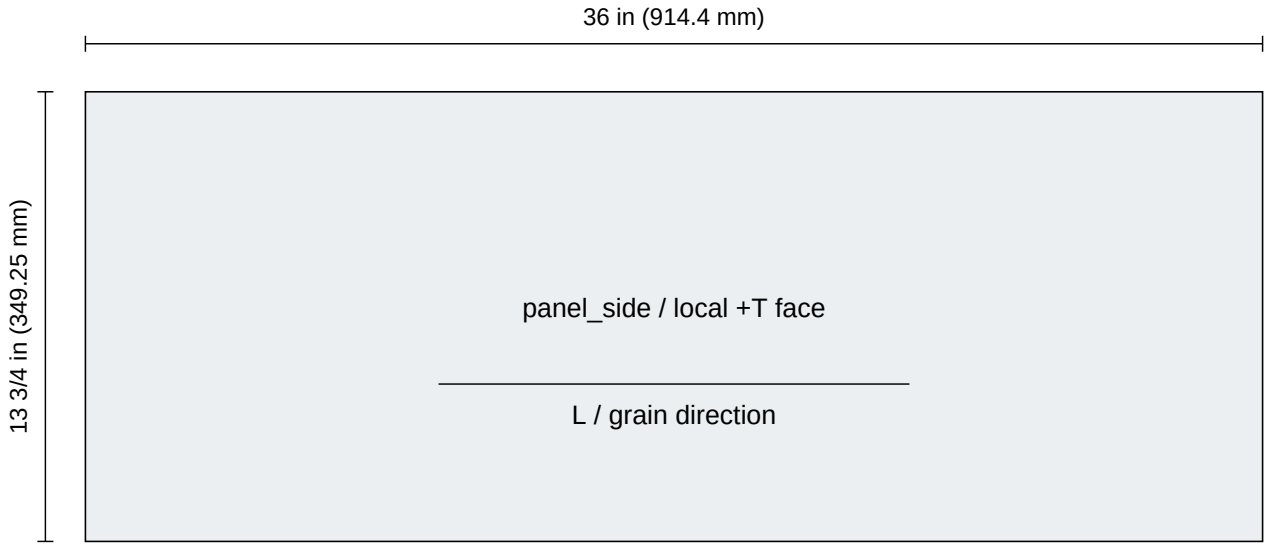
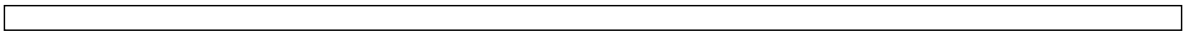
plywood_back	Example back plywood. Actual 1/4 in (6.35 mm). Nominal label: 1/4 in plywood (label only) (display only).
plywood_main	Example main plywood. Actual 3/4 in (19.05 mm). Nominal label: 3/4 in plywood (label only) (display only).
Cut process	Saw kerf 3.2 mm; rough allowance per edge 0 mm. No sufficient sheet count is asserted without a verified stock layout.

Measured part - panel_side

Physical members	side_left, side_right
Local length L	36 in (914.4 mm)
Local width W	13 3/4 in (349.25 mm)
Actual thickness T	3/4 in (19.05 mm)
Local convention	Centered origin. +X=L, +Y=W, +Z=T. Grain follows L where declared. This face diagram is nominal and uncompensated.



Thickness profile

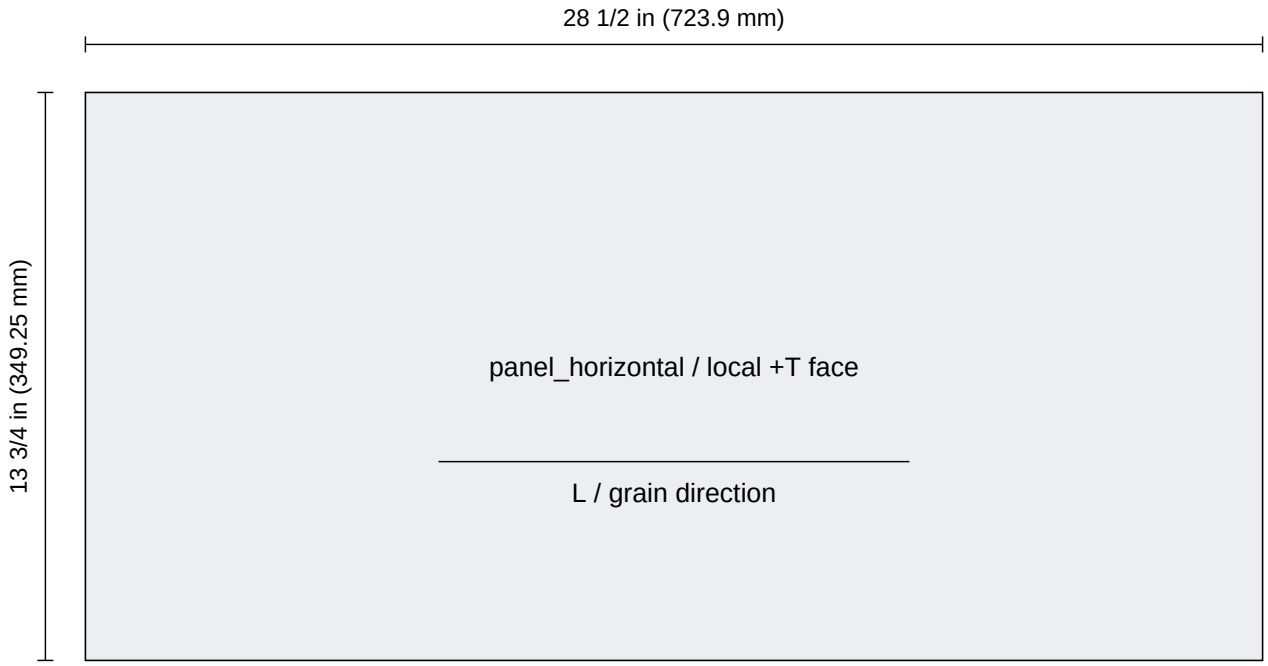


T = 3/4 in (19.05 mm)

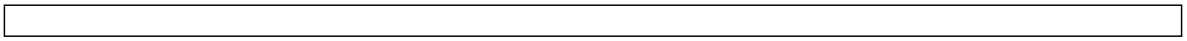
Companion drawing: drawings/panel_side-metric.pdf. Genuine cadgen A4 orthographic/isometric projections and native dimensions remain millimeters.

Measured part - panel_horizontal

Physical members	bottom, shelf_01, shelf_02, top
Local length L	28 1/2 in (723.9 mm)
Local width W	13 3/4 in (349.25 mm)
Actual thickness T	3/4 in (19.05 mm)
Local convention	Centered origin. +X=L, +Y=W, +Z=T. Grain follows L where declared. This face diagram is nominal and uncompensated.



Thickness profile

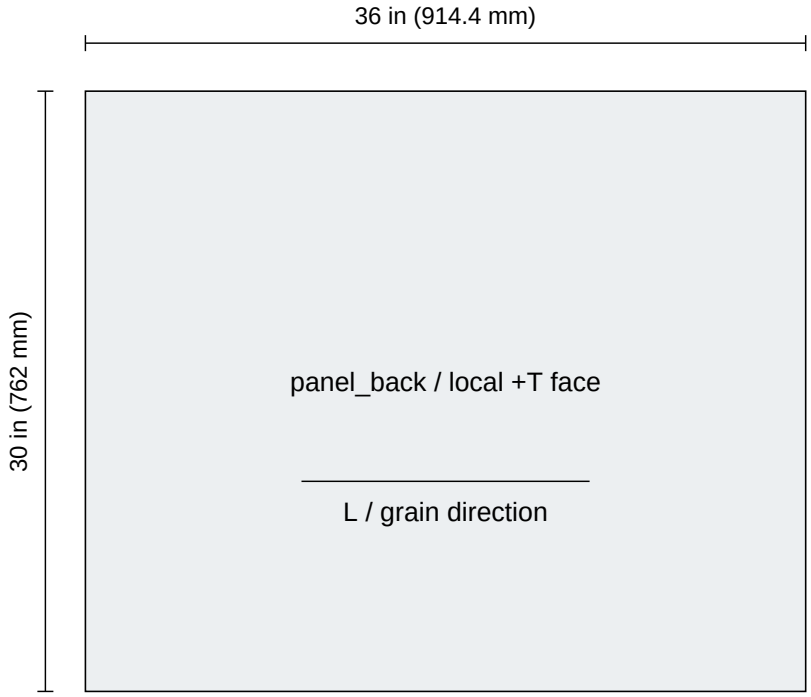


T = 3/4 in (19.05 mm)

Companion drawing: drawings/panel_horizontal-metric.pdf. Genuine cadgen A4 orthographic/isometric projections and native dimensions remain millimeters.

Measured part - panel_back

Physical members	back
Local length L	36 in (914.4 mm)
Local width W	30 in (762 mm)
Actual thickness T	1/4 in (6.35 mm)
Local convention	Centered origin. +X=L, +Y=W, +Z=T. Grain follows L where declared. This face diagram is nominal and uncompensated.



Thickness profile

T = 1/4 in (6.35 mm)

Companion drawing: drawings/panel_back-metric.pdf. Genuine cadgen A4 orthographic/isometric projections and native dimensions remain millimeters.

Construction state and revision details

Construction status	complete
Physical review	NOT REVIEWED - no actual build, service behavior or load capacity is certified.
Template	plywood_bookcase_v1 / 1.0.0
Resolved SHA-256	9eaedec6809341739d3e8bd1e525a3133250495cc8ba5947cf26fc3526cc0fef
Publication source	set_1761cfc031b0472a9404cbbc1180c6ce

The current STEP occurrence map is bound to the exact assembly hash. Rebuilds receive current selectors; old face/occurrence references are not manufacturing identity.

Recipe hardware and source review

Recipe	kreg720_19_05_14in_bookcase / 1.0.0
Required tools	kreg_kphj720_or_720pro, supplied_easy_set_bit_and_collar, square_drive_pocket_bit, drill_driver, clamps, panel_saw, square, measuring_tools, 18g_brad_nailer_1_25in_compatible
pocket_screw	24 required; Kreg #8 coarse-thread zinc Maxi-Loc pocket-hole screw; SML-C125-100; length 1 1/4 in (31.75 mm); 100 purchased in 1 pack(s). Price unknown.
back_brad	32 required; DEWALT indoor steel 18-gauge brad, 1-1/4 in length; DWBN18PP - select 1-1/4 in; length 1 1/4 in (31.75 mm); purchase quantity unknown. Price unknown.

All horizontal panels use the same marked local +T face. The end edge is the jig reference; the manufacturer jig defines hole angle and drilling depth.

Three guide stations per horizontal end are spaced 152.4 mm. At the supported depth, end offsets are 22.225 mm, within the manual's edge range.

Back brads are centered on each contacted carcass edge, with 25.4 mm end offsets and at most 180 mm between stations. This placement is authored, not a manufacturer shelf-load specification.

Hardware identification, edge alignment, joint coverage, dependency order and source consistency received a software technical review. Workholding, driver access, thin-back fastening and service behavior require a physical build review.

Before putting the bookcase into service, select and review tip-over restraint for the actual wall substrate and intended use. No wall anchor is inferred from geometry.

Recipe primary sources and limits

KPHJ720/720PRO Global manual, 06/2020 104909 V1, English pages 12-17

https://www.kregtool.com/on/demandware.static/-/Library-Sites-RefArchSharedLibrary/default/dw59d64a56/manuals/KPHJ720PRO_Global.pdf

Gauge actual stock, select the 3/4 range stop collar and 1-1/4 in screw; clamp joints. Panel guide stations: 5/8 to 1-1/2 in from edges and 6 to 8 in apart.

Kreg screw guide, updated June 29, 2026

https://learn.kregtool.com/learn/how-to-select-right-pocket-hole-screw/

Actual thickness governs selection. Coarse thread is preferred for plywood; zinc pocket screws are for indoor dry use.

Kreg / The Creative Mom Built In Bookcase

https://learn.kregtool.com/plans/built-in-bookcase/

Published method uses 1-1/4 in pocket screws and wood glue, with a separately attached 1/4 in plywood back, wood glue and 1-1/4 in brads. Original dimensions and quantities are not reused.

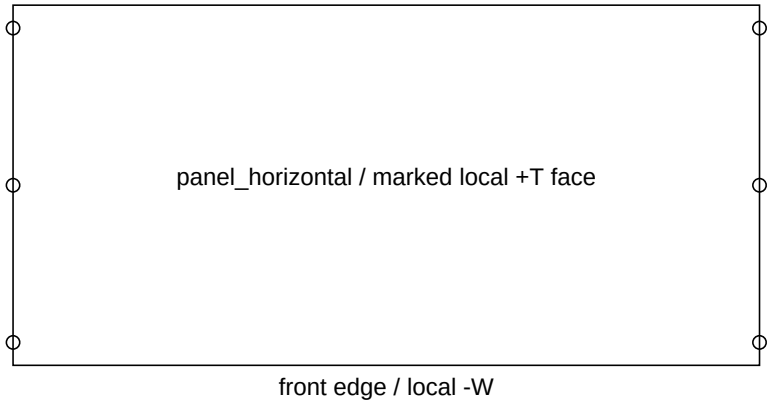
DEWALT DWBN18PP 18-gauge brad project pack

https://www.dewalt.com/en-us/product/dwbn18pp/18-gauge-brad-nail-project-pack

The indoor steel 18-gauge pack includes 1-1/4 in brads. Select that length; do not mix the other pack lengths into this recipe.

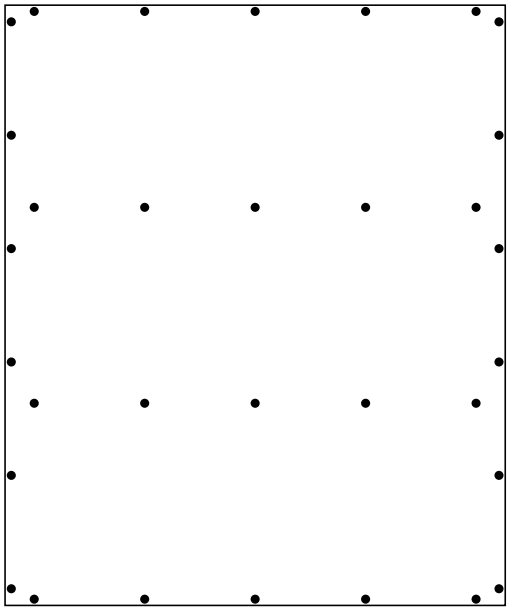
Operation reference diagrams

Horizontal panel: align the jig at the end-edge stations on marked local +T. Circles identify guide alignment references, not pocket-hole outlines. The manufacturer jig controls drilling depth and angle.



Front-edge station distances: 7/8 in (22.225 mm), 6 7/8 in (174.625 mm), 12 7/8 in (327.025 mm)

Applied back: drive the selected brads through its outer local +T face on the marked carcass-edge centerlines. This authored layout has no inferred shelf-load rating.



back / outside face: cabinet +X runs left; +Z runs up. Exact stations follow on the next page.

Back fastening reference coordinates

Coordinates below are cabinet-world millimeters: X is measured from the cabinet's left edge when viewed from the front; Z is measured from its bottom. When viewing the outside of back, cabinet +X runs left across the diagram. Align marked part IDs before fastening.

side_left / back	X centerline 9.525 mm; Z stations 25.4, 198.12, 370.84, 543.56, 716.28, 889 mm.
side_right / back	X centerline 752.475 mm; Z stations 25.4, 198.12, 370.84, 543.56, 716.28, 889 mm.
bottom / back	Z centerline 9.525 mm; X stations 44.45, 212.725, 381, 549.275, 717.55 mm.
shelf_01 / back	Z centerline 307.975 mm; X stations 44.45, 212.725, 381, 549.275, 717.55 mm.
shelf_02 / back	Z centerline 606.425 mm; X stations 44.45, 212.725, 381, 549.275, 717.55 mm.
top / back	Z centerline 904.875 mm; X stations 44.45, 212.725, 381, 549.275, 717.55 mm.

Brads are the separately sourced 1-1/4 in 18-gauge type, driven through the back's outer local +T face. Keep the thin back square, supported and flush; check all points for breakout. Station spacing is an authored geometric choice, not an approved load capacity.

Assembly sequence

1. cut_and_label

Parts: side_left, side_right, bottom, shelf_01, shelf_02, top, back

Cut and label every rough blank from this revision's cut schedule. Preserve grain along local L.

2. finish_to_size

Parts: side_left, side_right, bottom, shelf_01, shelf_02, top, back

Finish each rough blank to its exact finished L and W in the part schedule. Check actual thickness; rough allowance does not increase thickness.

3. set_jig

Parts: side_left, side_right, bottom, shelf_01, shelf_02, top, back

Check the measured main stock against the Kreg gauge's 3/4 range, set the supplied stop collar to that range, and securely clamp each panel. Use the manufacturer's jig procedure; no drill angle or depth is inferred from CAD.

4. drill_bottom

Parts: bottom

Mark bottom's local +T face. Align each end with the jig at the three listed stations measured from its front edge; drill the six pockets while the panel is still loose.

5. drill_shelf_01

Parts: shelf_01

Mark shelf_01's local +T face. Align each end with the jig at the three listed stations measured from its front edge; drill the six pockets while the panel is still loose.

6. drill_shelf_02

Parts: shelf_02

Mark shelf_02's local +T face. Align each end with the jig at the three listed stations measured from its front edge; drill the six pockets while the panel is still loose.

7. drill_top

Parts: top

Mark top's local +T face. Align each end with the jig at the three listed stations measured from its front edge; drill the six pockets while the panel is still loose.

8. dry_fit

Parts: side_left, side_right, bottom, shelf_01, shelf_02, top, back

Dry fit the carcass on a supported work surface. Compare each labeled panel center and each clear opening to this revision's drawing; check squareness before adhesive.

Assembly sequence

9. attach_left

Parts: side_left, bottom, shelf_01, shelf_02, top

Lay side_left on a supported work surface. Apply the selected glue per its label, clamp each horizontal panel in its drawn position, and seat the three specified pocket screws at each left end without overdriving.

Connections: attach_bottom_to_side_left, attach_shelf_01_to_side_left, attach_shelf_02_to_side_left, attach_top_to_side_left

10. attach_right

Parts: side_right, bottom, shelf_01, shelf_02, top

Fit side_right to the horizontal ends, keep the carcass square, clamp the joints and seat three specified screws per right end. Keep the front open for driver access.

Connections: attach_bottom_to_side_right, attach_shelf_01_to_side_right, attach_shelf_02_to_side_right, attach_top_to_side_right

11. attach_back

Parts: side_left, side_right, bottom, shelf_01, shelf_02, top, back

Support the square carcass face-down. Transfer the listed carcass-edge centerlines to back, apply wood glue per its label, clamp the back flush, and drive only the specified 1-1/4 in brads on those lines. Check the thin back and all edges for unintended breakout.

Connections: attach_side_left_to_back, attach_side_right_to_back, attach_bottom_to_back, attach_shelf_01_to_back, attach_shelf_02_to_back, attach_top_to_back

12. final_check

Parts: side_left, side_right, bottom, shelf_01, shelf_02, top, back

Allow adhesive to cure per its label. Check squareness, finished dimensions, clear openings, seated fasteners and breakout. Before use, arrange site-specific tip-over restraint and record the physical review; no shelf load rating is supplied.

Spec SHA-256 e44d9f09f593efb98fe8cc4da6505196e45b01f9e4012e8ba5e391ae3b35d409